

TrustXAI: An Explainable Ensemble Learning Framework for Trustworthy Multiclass Network Intrusion Detection

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Abstract

Modern network environments are increasingly threatened by sophisticated cyberattacks, creating an urgent need for intelligent and trustworthy Intrusion Detection Systems (IDS). Although machine learning techniques have significantly improved intrusion detection performance, their limited interpretability restricts their adoption in real-world cybersecurity applications. This paper presents TrustXAI, an Explainable Artificial Intelligence (XAI) framework for multiclass network intrusion detection using the CICIDS2017 dataset. The proposed framework evaluates two ensemble learning algorithms, namely Random Forest and Extreme Gradient Boosting (XGBoost), to classify multiple categories of network attacks. To improve transparency and model trustworthiness, SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) are integrated to provide both global and local explanations of model predictions. Experimental results demonstrate that the Random Forest classifier achieved the highest overall performance, attaining an accuracy of 99.94%, while XGBoost also delivered highly competitive classification results. Furthermore, the explainability analysis identified the most influential network traffic features affecting classification decisions, providing valuable insights into model behavior and improving analyst confidence. Overall, the proposed TrustXAI framework demonstrates that integrating Explainable Artificial Intelligence with high-performance ensemble learning models enhances the reliability, transparency, and practical applicability of intelligent cybersecurity systems.

Keywords— Explainable Artificial Intelligence (XAI), Cybersecurity, Intrusion Detection System (IDS), LIME, Machine Learning, Random Forest, SHAP, XGBoost , CICIDS2017.

1. Introduction

The rapid growth of digital technologies, cloud computing, the Internet of Things (IoT), and smart infrastructures has significantly increased the complexity of modern network environments. As organizations become increasingly dependent on interconnected systems, cyberattacks have evolved in both frequency and sophistication, posing serious threats to data confidentiality, system availability, and critical infrastructure. Consequently, Intrusion Detection Systems (IDS) have become an essential component of modern cybersecurity architectures [11], [25].

Traditional signature-based intrusion detection techniques are effective in identifying previously known attacks; however, they often fail to detect novel or sophisticated threats. To overcome these limitations, machine learning (ML) techniques have been widely adopted due to their ability to automatically learn complex patterns from network traffic and detect both known and unknown attacks. Recent studies have demonstrated that ensemble learning algorithms such as Random Forest and Extreme Gradient Boosting (XGBoost) achieve high detection accuracy across various intrusion detection datasets [1], [2], [11], [23].

Despite their impressive predictive performance, most machine learning models remain difficult to interpret. Security analysts frequently require understandable explanations for model predictions before trusting automated decisions, especially in mission-critical environments. This lack of transparency has motivated increasing research interest in Explainable Artificial Intelligence (XAI), which aims to improve the interpretability and trustworthiness of machine learning systems without significantly compromising their predictive capabilities [13], [24].

Among the most widely adopted XAI techniques, SHapley Additive exPlanations (SHAP) provides global feature importance analysis based on cooperative game theory, whereas Local Interpretable Model-Agnostic Explanations (LIME) explains individual predictions through locally interpretable surrogate models. The integration of these complementary approaches enables security professionals to understand both overall model behavior and case-specific prediction decisions [3], [4].

Motivated by these challenges, this paper proposes TrustXAI, an Explainable Artificial Intelligence framework for multiclass network intrusion detection using the CICIDS2017 dataset [5]. The proposed framework evaluates two powerful ensemble learning models, Random Forest and XGBoost, while integrating SHAP and LIME to enhance model transparency and interpretability. Experimental evaluation demonstrates that the proposed framework achieves excellent classification performance while providing meaningful explanations that improve trust in machine learning-based intrusion detection systems.

The main contributions of this paper are summarized as follows:

1. A multiclass intrusion detection framework based on Random Forest and XGBoost was developed and evaluated using the CICIDS2017 dataset.
2. Explainable Artificial Intelligence techniques, namely SHAP and LIME, were integrated to provide both global and local model interpretability, improving the transparency and trustworthiness of machine learning-based intrusion detection.
3. A comprehensive comparative analysis between Random Forest and XGBoost was conducted using multiple evaluation metrics, including Accuracy, Precision, Recall, and F1-score.
4. Experimental results demonstrated that the Random Forest classifier achieved the highest overall classification performance, while the integration of SHAP and LIME significantly enhanced model transparency and interpretability.

2. Related Work

2.1 Machine Learning for Intrusion Detection

Machine learning has become one of the most widely adopted approaches for network intrusion detection due to its ability to identify complex attack patterns from large-scale network traffic. Unlike traditional signature-based methods, machine learning algorithms are capable of detecting both known and previously unseen attacks by learning discriminative features directly from data [11], [14], [25]. Among the numerous supervised learning algorithms, ensemble learning techniques such as Random Forest (RF) and Extreme Gradient Boosting (XGBoost) have consistently demonstrated excellent performance in cybersecurity applications because of their robustness, high predictive accuracy, and resistance to overfitting [1], [2], [12], [23].

Recent research has shown that XGBoost frequently achieves superior classification performance on benchmark intrusion detection datasets owing to its gradient boosting strategy and effective handling of complex nonlinear relationships [2], [23]. Similarly, Random Forest remains a reliable baseline model because of its simplicity, stability, and strong generalization capability across different attack categories [1], [11]. Consequently, these two algorithms have become popular choices for intelligent intrusion detection systems [12], [23].

2.2 Explainable Artificial Intelligence in Cybersecurity

Although modern machine learning algorithms have significantly improved intrusion detection performance, many of these models operate as black-box systems, making it difficult for cybersecurity analysts to understand the reasoning behind their predictions. In security-critical environments, decision transparency is essential because analysts must verify whether a detected event represents a genuine cyberattack or a false alarm.

Consequently, Explainable Artificial Intelligence (XAI) has emerged as an important research direction for improving the interpretability and trustworthiness of intelligent cybersecurity systems [13], [24].

Among the various XAI techniques, SHapley Additive exPlanations (SHAP) has become one of the most widely adopted approaches for global model interpretation by measuring the contribution of each feature to the final prediction [4]. In contrast, Local Interpretable Model-Agnostic Explanations (LIME) focuses on explaining individual predictions by constructing simple surrogate models around specific instances [3]. The combination of SHAP and LIME provides complementary insights, allowing researchers to analyze both overall model behavior and instance-level decision-making [3], [4], [24].

Recent studies have demonstrated that integrating XAI techniques into intrusion detection systems improves analyst confidence, facilitates model validation, and enhances decision transparency [13], [16], [24]. These advantages make SHAP and LIME particularly suitable for cybersecurity applications where explainability is as important as predictive performance [16], [24].

2.3 Research Gap

Despite the considerable progress achieved in machine learning-based intrusion detection, several limitations remain. First, many existing studies focus primarily on maximizing classification accuracy while paying limited attention to model interpretability [11], [13]. Second, a large proportion of previous research evaluates only a single machine learning algorithm, making it difficult to understand the relative strengths of different ensemble methods [12], [23]. Third, only a limited number of studies combine both global and local explainability techniques within the same intrusion detection framework [13], [16], [24]. Furthermore, comprehensive comparisons between Random Forest and XGBoost using both SHAP and LIME remain relatively scarce [12], [16].

These limitations highlight the need for a unified framework that simultaneously delivers high predictive performance, transparent decision-making, and comparative explainability analysis for multiclass intrusion detection. Therefore, this study addresses these limitations by proposing an explainable intrusion detection framework that combines two ensemble learning models with complementary XAI techniques to improve both predictive performance and transparency.

3. Proposed TrustXAI Framework

3.1 Framework Overview

The proposed TrustXAI framework was designed to develop an accurate and interpretable multiclass intrusion detection system capable of detecting various network attacks while providing transparent explanations for every prediction. Unlike conventional machine learning-based intrusion detection systems that primarily focus on predictive performance, TrustXAI integrates Explainable Artificial Intelligence (XAI) techniques into the entire detection pipeline to improve both model transparency and user trust [3], [4], [13], [24].

The framework consists of several sequential stages, including data acquisition, preprocessing, feature engineering, model training, performance evaluation, and explainability analysis. Two ensemble learning algorithms, namely Random Forest and XGBoost, were implemented and compared using identical datasets and evaluation metrics [1], [2]. Finally, SHAP and LIME were employed to explain both global model behavior and individual prediction decisions [3], [4]

Figure.1 illustrates the overall architecture of the proposed TrustXAI framework.

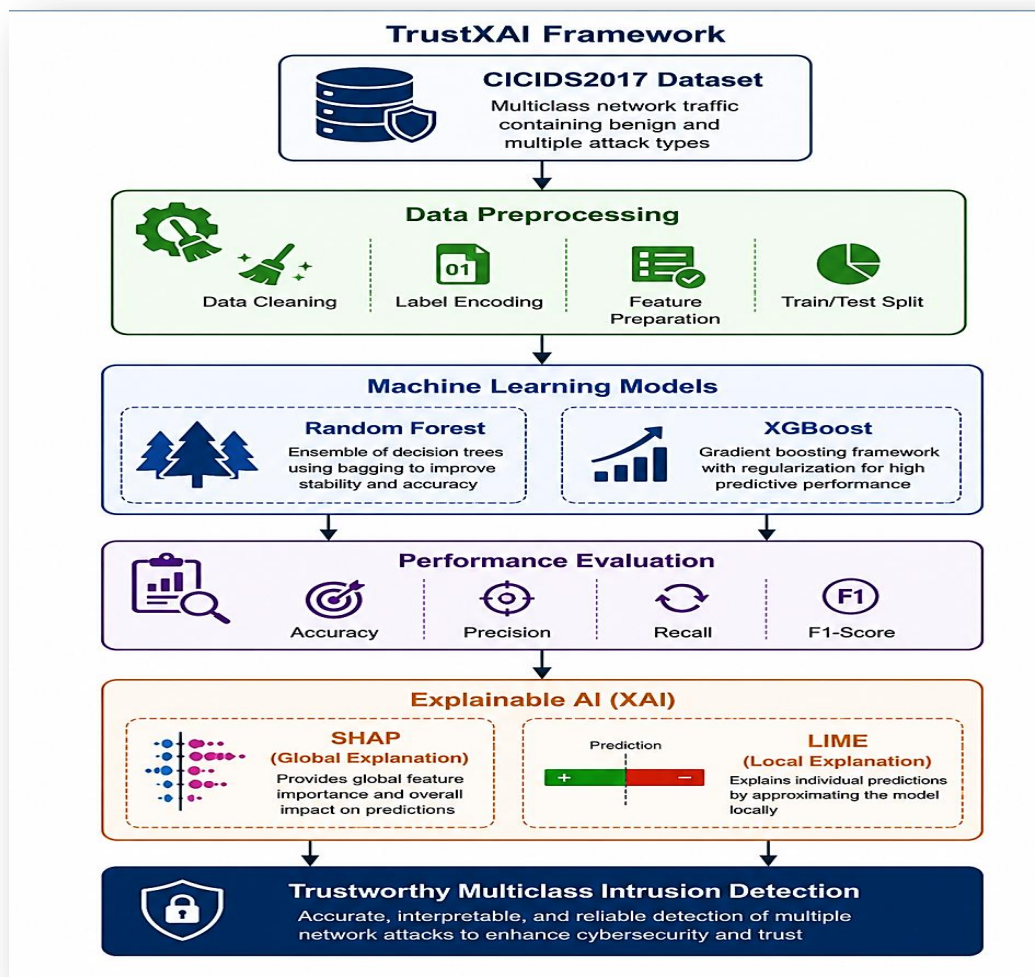


Figure 1 illustrates the overall workflow of the proposed TrustXAI framework. The framework begins with the acquisition of the CICIDS2017 dataset [5], followed by a comprehensive data preprocessing stage, including data cleaning, label encoding, and feature preparation. The processed dataset is then divided into training and testing subsets before being used to train the Random Forest and XGBoost classifiers [1], [2]. After model training, performance is evaluated using standard classification metrics, including Accuracy, Precision, Recall, and F1-score. Finally, Explainable Artificial Intelligence techniques, namely SHAP and LIME, are applied to provide global and local explanations of model predictions, improving transparency, interpretability, and trustworthiness in cybersecurity decision-making [3], [4], [13]

3.2 Dataset Description

The experiments conducted in this study were performed using the CICIDS2017 dataset, one of the most widely adopted benchmark datasets for network intrusion detection research [5]. The dataset was generated by the Canadian Institute for Cybersecurity (CIC) and contains realistic network traffic collected under normal user activities as well as multiple categories of cyberattacks.

CICIDS2017 includes a comprehensive set of network flow features describing communication behavior between network hosts. The dataset contains eighty extracted traffic features representing statistical characteristics such as packet length, flow duration, protocol information, packet rates, and header attributes. These features enable machine learning algorithms to distinguish normal traffic from malicious activities effectively [5].

In this research, fifteen traffic classes were considered, including one benign class and fourteen attack categories, enabling the development of a realistic multiclass intrusion detection model.

3.3 Data Preprocessing

Data preprocessing plays a crucial role in improving the quality of machine learning models by ensuring that the input data are clean, consistent, and suitable for training [19], [21]. The CICIDS2017 dataset contains network traffic records with multiple numerical features extracted from bidirectional flows. Before model development, several preprocessing operations were performed to improve data quality and reduce potential sources of bias.

Initially, duplicated records were removed to eliminate redundant observations that could negatively influence the learning process. Missing and invalid values were carefully inspected and appropriately handled to ensure data consistency. Furthermore, categorical attack labels were transformed into numerical class identifiers to facilitate multiclass classification using supervised machine learning algorithms [5], [21].

Finally, the processed dataset was divided into training and testing subsets using a stratified sampling strategy in order to preserve the original class distribution and guarantee a fair evaluation of model performance [21]

Table 1. Distribution of Network Traffic Classes after Data Preprocessing

Class ID	Attack Category	Number of Samples
0	Benign	2,271,320
1	Bot	1,956
2	DDoS	128,025
3	DoS GoldenEye	10,293
4	DoS Hulk	230,124
5	DoS Slowhttptest	5,499
6	DoS Slowloris	5,796
7	FTP-Patator	7,935
8	Heartbleed	11
9	Infiltration	36
10	PortScan	158,804
11	SSH-Patator	5,897
12	Web Attack – Brute Force	1,507
13	Web Attack – SQL Injection	21
14	Web Attack – XSS	652

3.4 Feature Engineering

Feature engineering is an essential step in machine learning because the quality of extracted features directly influences model performance [19], [21]. The CICIDS2017 dataset provides eighty numerical traffic features representing different characteristics of network communication, including packet statistics, flow duration, protocol behavior, header information, and transmission rates [5].

Instead of manually selecting a limited subset of variables, this study utilized the complete set of eighty extracted traffic features to preserve all available information. During model interpretation, SHAP and LIME were subsequently employed to determine the relative importance of these features and identify the network attributes that contributed most significantly to attack classification decisions [3], [4]

3.5 Random Forest Classifier

Random Forest (RF) is a supervised ensemble learning algorithm that constructs multiple decision trees during the training phase and combines their predictions using majority voting for classification tasks. The algorithm reduces overfitting by introducing randomness in both feature selection and bootstrap sampling, resulting in improved generalization performance [1].

In this study, Random Forest was employed as one of the classifiers for multiclass intrusion detection. The model was trained using the processed CICIDS2017 dataset [5] and evaluated using Accuracy, Precision, Recall, and F1-score. Owing to its robustness and ability to handle high-dimensional data, Random Forest provides reliable performance for cybersecurity classification problems while maintaining relatively low computational complexity [1], [12]

The prediction of Random Forest can be expressed as:

$$\mathbf{y}^{\wedge} = \mathbf{mode}\{\mathbf{T1}(x), \mathbf{T2}(x), \dots, \mathbf{Tn}(x)\}$$

Where $\mathbf{T}_I(x)$ represents the prediction of the i^{th} decision tree, while the final class is determined by majority voting

3.6 Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) is an advanced ensemble learning algorithm based on gradient boosting decision trees. Unlike bagging-based approaches, XGBoost constructs trees sequentially, where each new tree attempts to minimize the prediction errors produced by previous trees. The algorithm incorporates regularization mechanisms that improve model generalization while reducing overfitting [2].

Due to its high predictive capability and computational efficiency, XGBoost has become one of the most widely adopted machine learning algorithms in cybersecurity research [2], [23]. In this study, XGBoost was trained using the same training dataset as the Random Forest classifier to ensure a fair and consistent comparative evaluation. The performance of XGBoost was subsequently compared with Random Forest using the same evaluation metrics, including Accuracy, Precision, Recall, and F1-score, as presented in the Experimental Results and Discussion section

The prediction model can be formulated as :

$$y_i = \sum_{k=1}^K f_k(x_i)$$

3.7 Explainable Artificial Intelligence using SHAP

Explainable Artificial Intelligence (XAI) has become an essential component of trustworthy machine learning systems, particularly in cybersecurity applications where understanding model decisions is critical [13], [24]. Among various XAI techniques, SHapley Additive exPlanations (SHAP) provides a theoretically sound framework for interpreting complex machine learning models by assigning contribution scores to each input feature [4].

SHAP is based on cooperative game theory and calculates the marginal contribution of every feature to the final prediction. Unlike traditional feature importance methods, SHAP explains both the magnitude and direction of each feature's influence, allowing security analysts to understand why a specific network flow is classified as benign or malicious [4].

In this study, SHAP was applied to the trained Random Forest model to provide global explanations of model behavior. The generated SHAP summary plot highlights the most influential network traffic features and demonstrates how feature values affect the classification outcome across all attack categories

The SHAP value for feature i is computed as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

Where:

- F represents the complete feature set.
- S is a subset of features.
- $f(\cdot)$ denotes the prediction function.
- ϕ_i measures the contribution of feature i .

Figure 2 presents the SHAP summary plot generated from the trained XGBoost model.

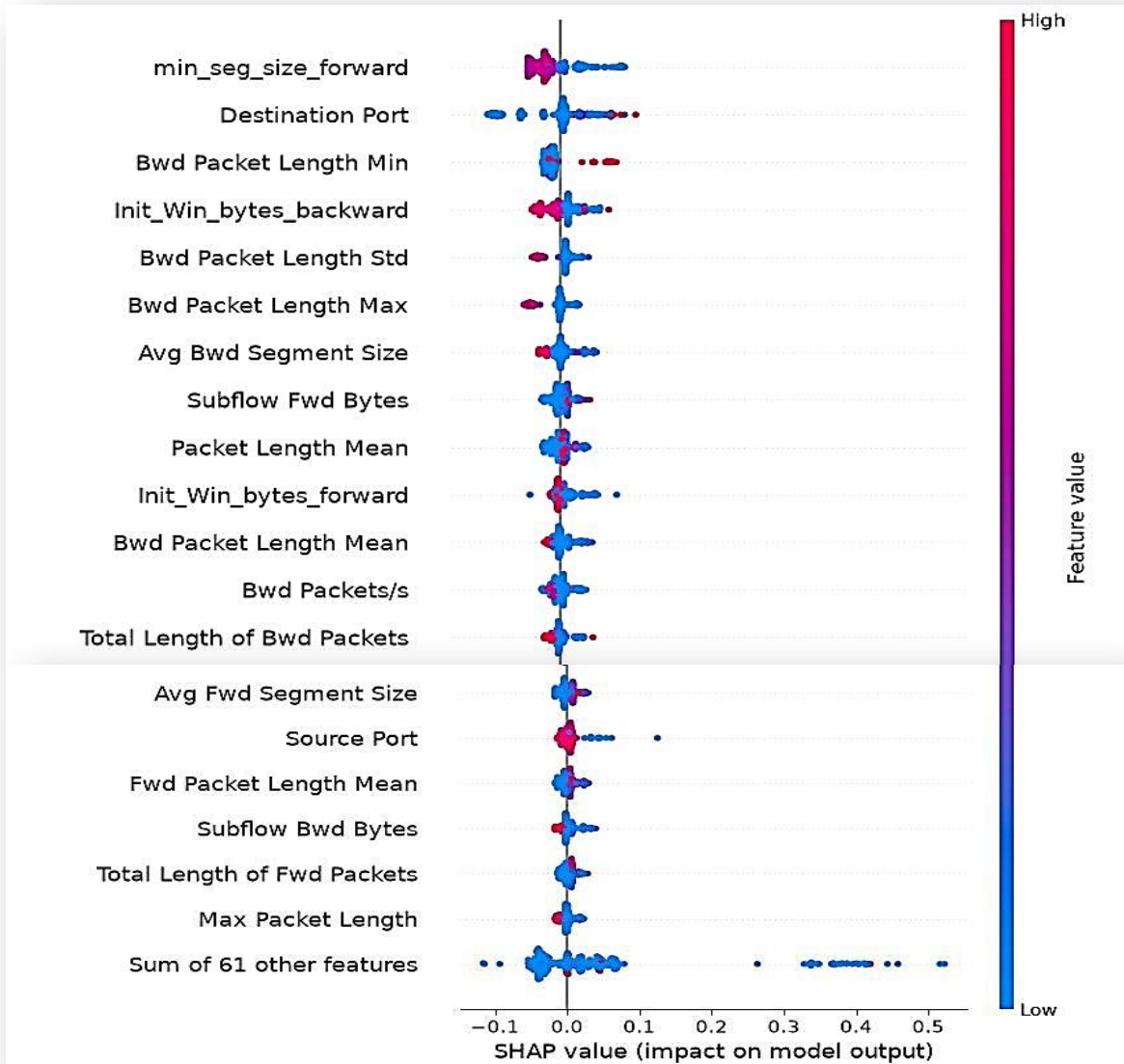


Figure 2. SHAP Summary Plot for Global Feature Importance

As illustrated in Figure 2, features located at the top of the summary plot have the greatest overall impact on model predictions. Each point represents an individual network flow, while the color indicates the corresponding feature value. Red points denote high feature values, whereas blue points indicate lower values.

The distribution of SHAP values demonstrates how different traffic characteristics influence attack classification decisions. Features with large positive or negative SHAP values contribute significantly to the final prediction, providing valuable insights into the internal behavior of the Random Forest model. Such transparency enables cybersecurity

analysts to validate automated decisions and increases confidence in the proposed intrusion detection framework

3.8 Local Explainability using LIME

While SHAP provides global explanations describing the overall behavior of the machine learning model, Local Interpretable Model-Agnostic Explanations (LIME) focuses on explaining individual predictions [3]. LIME approximates the behavior of a complex machine learning model around a single observation using an interpretable surrogate model.

In cybersecurity applications, local explanations are particularly valuable because security analysts frequently investigate individual alerts rather than the entire dataset. LIME identifies the most influential features responsible for a specific prediction, allowing analysts to understand why a network connection has been classified as a particular attack category [3], [24].

Figure 3 illustrates the LIME explanation generated for one network traffic instance.

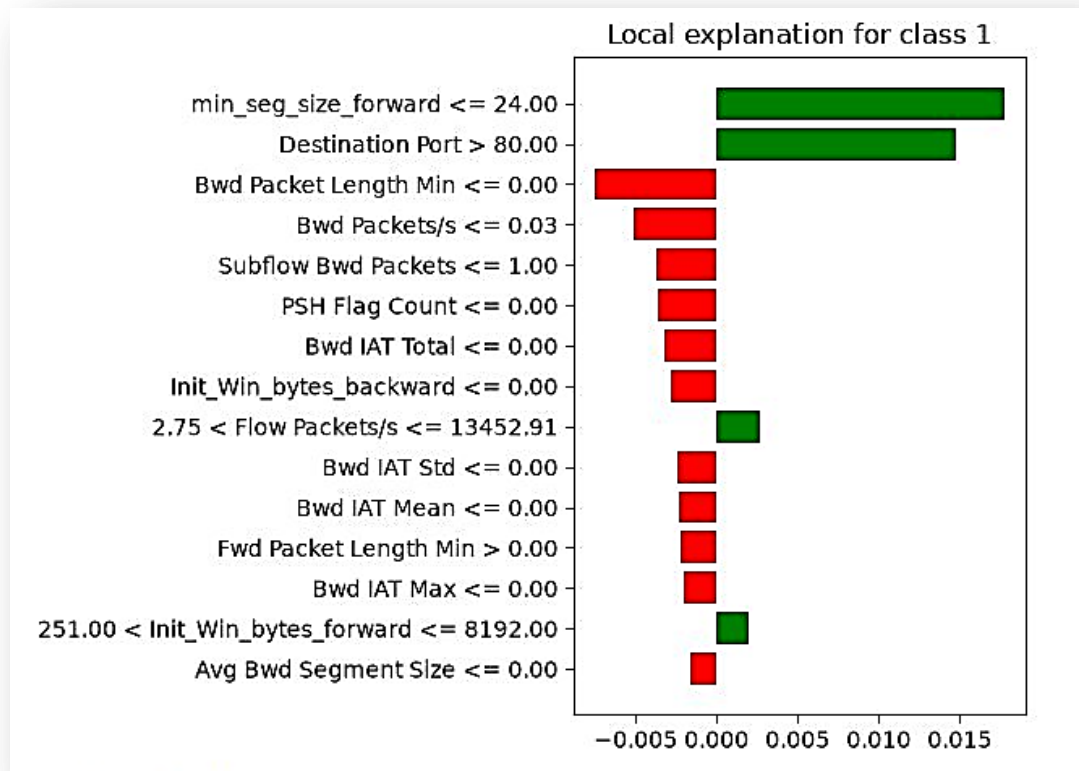


Figure 3. Local Prediction Explanation using LIME

Figure 3 demonstrates the local interpretation generated by LIME for an individual network traffic record. The visualization highlights both positive and negative feature contributions affecting the predicted attack category. Features displayed on the right support the predicted class, whereas features shown on the left decrease the confidence of that prediction.

Compared with SHAP, which explains the overall model behavior, LIME provides instance-level explanations that enable analysts to investigate specific alerts. The combination of SHAP and LIME therefore offers complementary perspectives, significantly improving the transparency and trustworthiness of machine learning-based intrusion detection systems [3], [4], [24]

4. Experimental Results and Discussion

4.1 Experimental Setup

All experiments were conducted using Python within the Anaconda environment and implemented through Jupyter Notebook. The machine learning models were developed using the Scikit-learn and XGBoost libraries [2], [21], while SHAP and LIME were employed to provide explainability analysis [3], [4]. The CICIDS2017 dataset was divided into training and testing subsets using a stratified train-test split strategy to ensure a balanced evaluation across all attack categories [5], [21].

To evaluate the effectiveness of the proposed framework, four widely adopted classification metrics were considered: Accuracy, Precision, Recall, and F1-score. These metrics provide a comprehensive assessment of both predictive performance and classification reliability for multiclass intrusion detection [11], [21].

4.2 Performance Evaluation

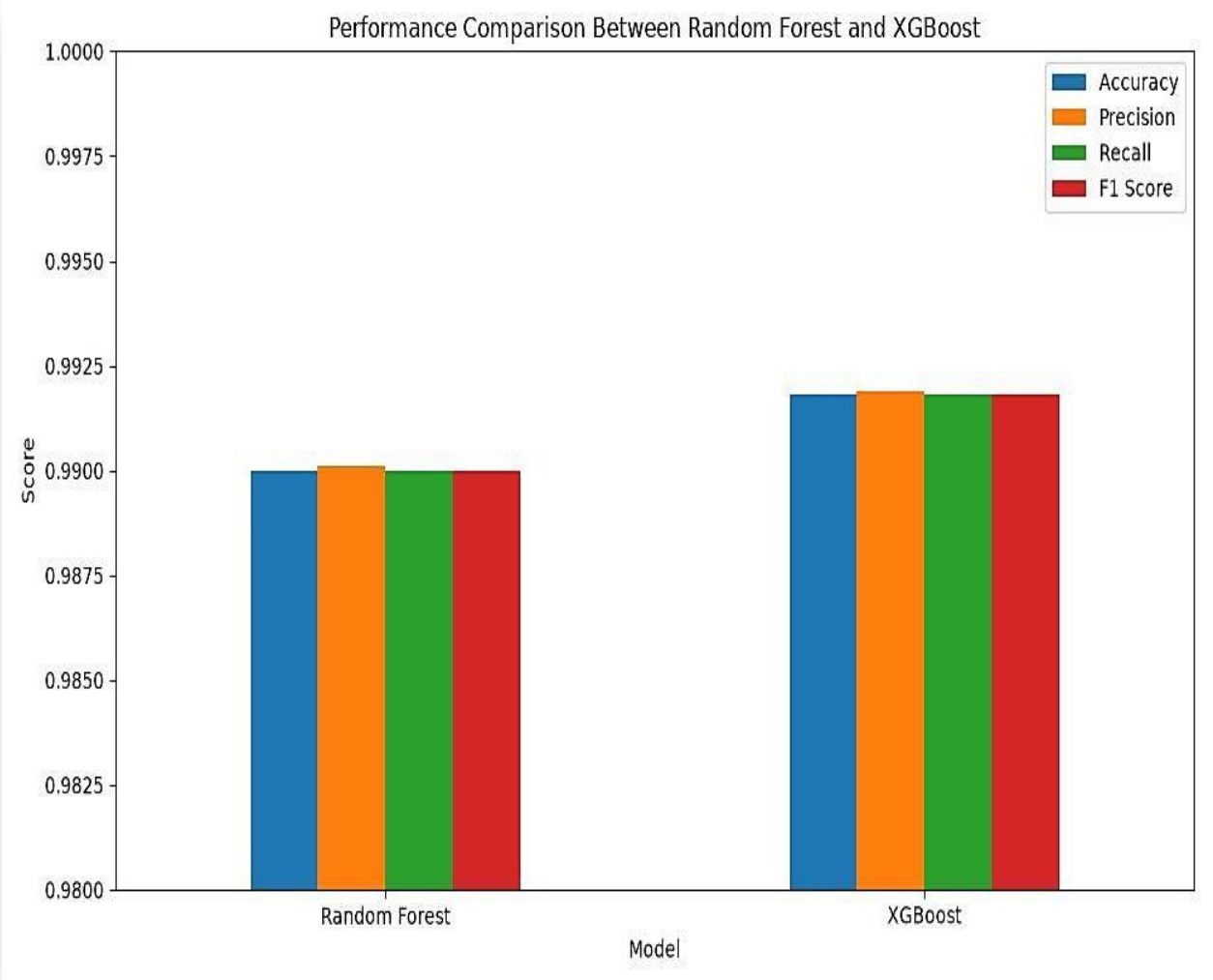
Both ensemble learning algorithms demonstrated excellent predictive capability on the CICIDS2017 dataset. However, the Random Forest classifier achieved the highest overall performance across all evaluation metrics, indicating its effectiveness in multiclass intrusion detection, while XGBoost also produced highly competitive results [1], [2].

Table 2. Performance Comparison between Random Forest and XGBoost

Model	Accuracy	Precision	Recall	F1-score
Random Forest	99.94%	99.94%	99.94%	99.94%
XGBoost	99.18%	99.19%	99.18%	99.18%

As shown in Table 2, the Random Forest classifier achieved the highest overall classification performance, reaching an accuracy of 99.94%, while maintaining equally high Precision, Recall, and F1-score values. These findings demonstrate the effectiveness of the proposed TrustXAI framework for accurately distinguishing multiple categories of network attacks with minimal classification errors. Although XGBoost also achieved excellent classification performance, Random Forest consistently produced superior results across all evaluation metrics, making it the most effective model within the proposed framework.

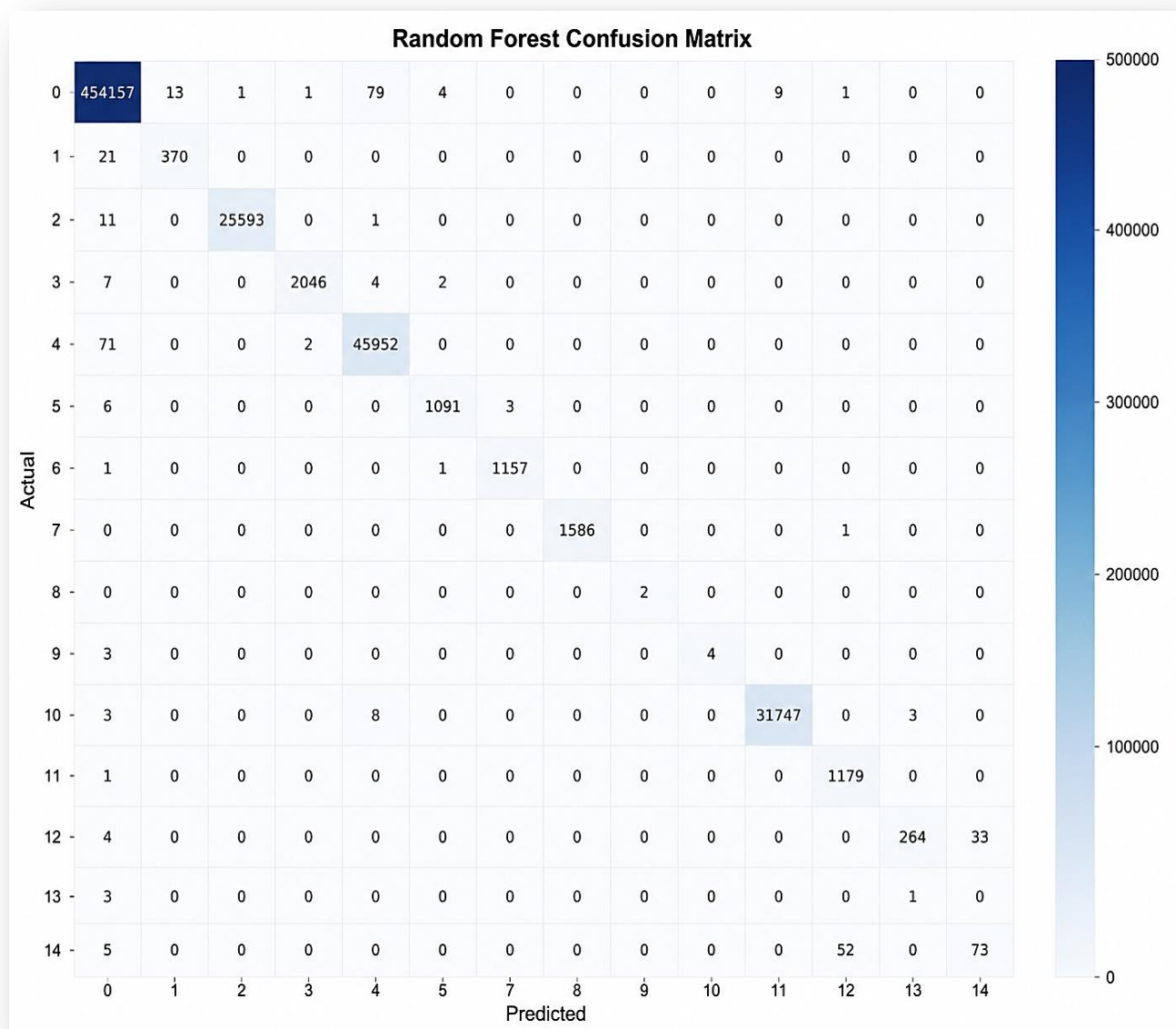
Figure 4. Performance Comparison Between Random Forest and XGBoost



The confusion matrix confirms the strong classification capability of the Random Forest model.

Figure 5. Random Forest Confusion Matrix

he confusion matrix provides a detailed evaluation of the classification performance achieved by the Random Forest model. It illustrates the relationship between the actual and predicted classes, allowing the identification of correctly classified instances as well as potential misclassifications [1], [21].

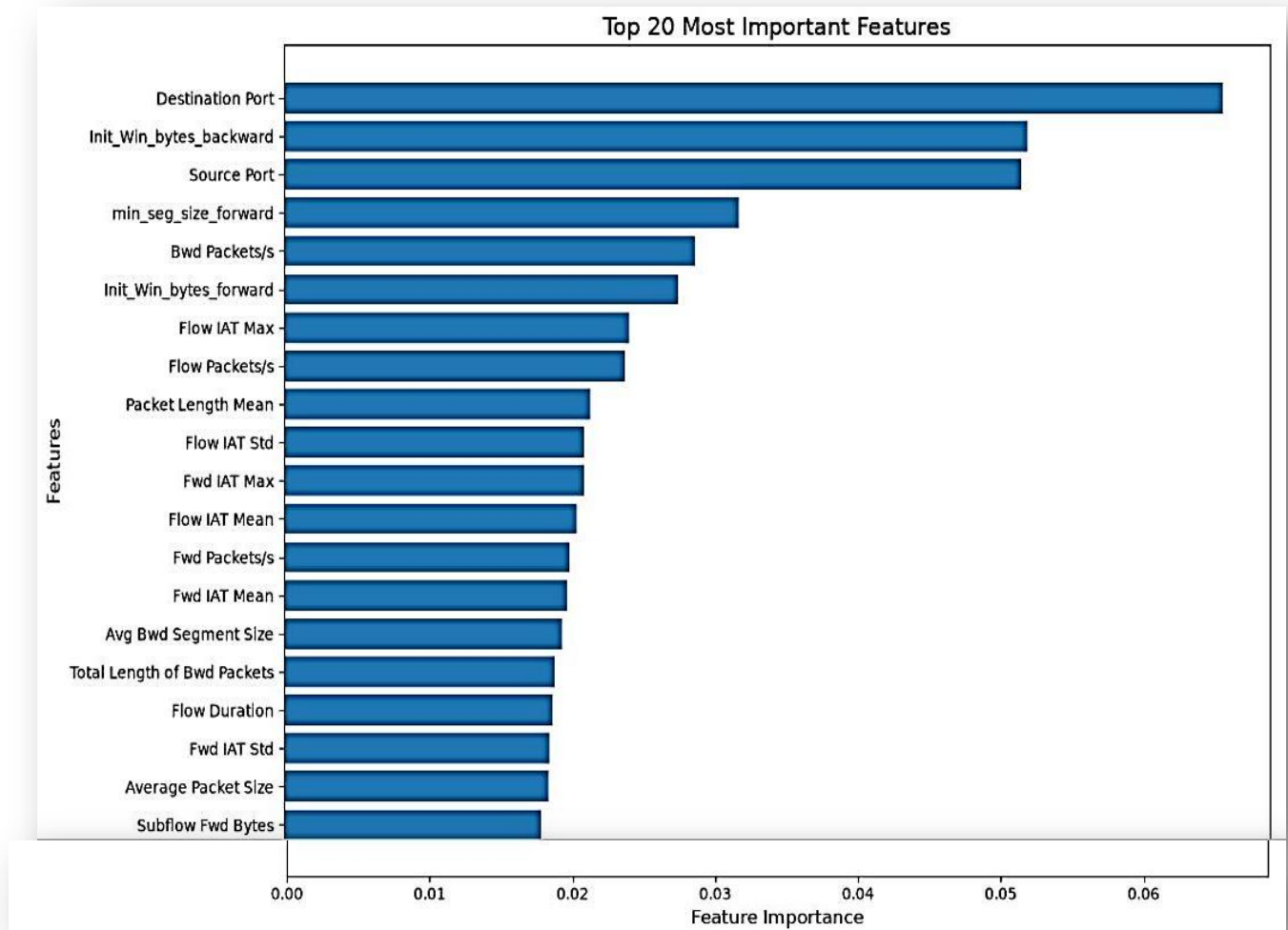


As illustrated in Figure 5, the majority of network traffic instances were correctly classified across all attack categories, while only a small number of misclassifications occurred in minority classes. The dominant diagonal values indicate excellent predictive performance and demonstrate the ability of the Random Forest classifier to distinguish between benign traffic and multiple attack categories with high reliability. These results

further confirm the robustness and effectiveness of the proposed TrustXAI framework for multiclass intrusion detection.

Figure 6. Random Forest Feature Importance Ranking

Understanding which network traffic features contribute most significantly to the classification process is essential for improving model interpretability. The Random Forest algorithm estimates feature importance based on the contribution of each feature to reducing impurity across all decision trees [1].



As shown in Figure 6, only a limited number of network traffic features contribute substantially to the classification process, whereas many remaining features have relatively minor influence. The highest-ranked features play a critical role in distinguishing between normal network traffic and malicious activities. This analysis complements the SHAP and LIME explanations by providing an additional global interpretation of the model and increasing confidence in the decision-making process.

4.3 Global Model Interpretation using SHAP

To evaluate the interpretability of the proposed intrusion detection framework, SHAP (SHapley Additive exPlanations) was applied to the trained Random Forest model [4]. The generated SHAP summary plot provides a global interpretation of model behavior by ranking network traffic features according to their overall contribution to the classification process. This visualization enables cybersecurity analysts to identify the most influential variables responsible for distinguishing benign traffic from different attack categories.

The SHAP analysis revealed that only a relatively small subset of network traffic features contributed substantially to the final classification process, while many remaining features exhibited limited influence. Features with the highest SHAP values played the most significant role in distinguishing between benign and malicious traffic. Furthermore, the distribution of SHAP values demonstrated how high and low feature values influenced the model differently according to the learned decision boundaries. These global explanations provide valuable insights into the internal behavior of the Random Forest model, improving transparency and enabling cybersecurity professionals to better understand and validate automated classification decisions [4], [24]

4.4 Local Prediction Interpretation using LIME

While SHAP provides global explanations of the model's overall behavior, LIME focuses on interpreting individual predictions [3]. This local explanation is particularly valuable in cybersecurity, where analysts often investigate specific network events and security alerts.

The LIME analysis identified the features that positively and negatively influenced individual prediction outcomes. Features with positive contributions increased the confidence of the predicted attack category, whereas negatively weighted features reduced that confidence. These instance-level explanations complement the global insights obtained from SHAP, providing a more comprehensive understanding of the model's decision-making process and improving analyst trust in the proposed TrustXAI framework [3], [4], [24].

4.5 Comparative Discussion

The experimental evaluation demonstrates that both Random Forest and XGBoost provide excellent classification performance for multiclass intrusion detection on the CICIDS2017 dataset [5]. However, the Random Forest classifier achieved the highest overall predictive performance, obtaining superior Accuracy, Precision, Recall, and F1-score values. This superior performance can be attributed to the robustness of the Random Forest ensemble learning strategy, its effective handling of high-dimensional

network traffic features, and its strong generalization capability across multiple attack categories [1].

From an explainability perspective, the integration of SHAP and LIME significantly enhanced the transparency of the proposed framework. SHAP provided comprehensive global insights into feature importance across the entire dataset, while LIME generated detailed explanations for individual network traffic instances. The combination of these complementary Explainable Artificial Intelligence techniques enables cybersecurity analysts to understand both the overall behavior of the model and the reasoning behind specific classification decisions [3], [4], [13], [24].

Overall, the proposed TrustXAI framework successfully combines high predictive performance with transparent decision-making, making it a reliable and practical solution for real-world cybersecurity environments where both detection accuracy and model interpretability are essential.

5. Conclusion

This paper presented TrustXAI, an Explainable Artificial Intelligence (XAI) framework for multiclass network intrusion detection using the CICIDS2017 dataset. The proposed framework integrates two ensemble learning algorithms, namely Random Forest and Extreme Gradient Boosting (XGBoost), together with SHAP and LIME to achieve both high predictive performance and improved model interpretability.

Experimental results demonstrated that both ensemble learning models achieved outstanding classification performance. However, the Random Forest classifier consistently outperformed XGBoost, achieving an overall accuracy of 99.94% while maintaining equally high Precision, Recall, and F1-score values. In addition, the integration of SHAP and LIME significantly improved model transparency by providing comprehensive global and local explanations of prediction decisions, enabling a better understanding of the model's behavior and increasing analyst confidence.

The obtained results confirm that the proposed TrustXAI framework successfully addresses the research gap identified in the literature by combining high predictive performance with comprehensive model interpretability within a unified intrusion detection framework.

Overall, the proposed TrustXAI framework demonstrates that combining high-performance ensemble learning with Explainable Artificial Intelligence can significantly enhance the accuracy, transparency, reliability, and practical applicability of intelligent intrusion detection systems for modern cybersecurity environments. The framework therefore provides a practical foundation for developing trustworthy, interpretable, and

high-performance intrusion detection systems capable of supporting future cybersecurity operations

6. Future Work

Although the proposed TrustXAI framework achieved promising results for multiclass intrusion detection, several research directions remain open for future investigation.

First, future studies may evaluate additional deep learning architectures, including Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), Convolutional Neural Networks (CNN), Transformer-based models, and Graph Neural Networks (GNNs), to further improve intrusion detection performance on increasingly complex and evolving network traffic.

Second, the proposed framework can be validated using additional benchmark cybersecurity datasets, such as CSE-CIC-IDS2018, UNSW-NB15, BoT-IoT, and TON_IoT, to investigate its robustness and generalization capability across diverse network environments and attack scenarios.

Furthermore, future research may incorporate real-time network traffic monitoring, online learning mechanisms, and incremental model updating to enable continuous adaptation against emerging cyber threats without requiring complete model retraining.

Another promising research direction involves integrating uncertainty estimation and prediction confidence scoring into the Explainable Artificial Intelligence framework. Providing confidence levels alongside model explanations could further assist cybersecurity analysts in prioritizing security alerts, reducing false positives, and improving incident response efficiency.

Future work may also investigate the integration of the proposed TrustXAI framework with Security Information and Event Management (SIEM) platforms to evaluate its effectiveness in real-world cybersecurity environments and support automated security monitoring and incident response.

Finally, future work may focus on developing an interactive cybersecurity decision-support platform that combines SHAP, LIME, and threat intelligence into a unified visualization dashboard for Security Operations Centers (SOCs). Such a platform would improve the practical deployment of Explainable Artificial Intelligence and support more transparent, trustworthy, and efficient cybersecurity operations.

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